An Ionization-Aware Field-Aligned Chromospheric Column Model for Conduction-Driven Evaporation

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Abstract

Global coronal models commonly replace the dynamic chromosphere and its thin transition region with a simplified lower boundary, limiting their ability to represent time-dependent mass loading by conductive heating. We present a one-dimensional field-aligned chromospheric column designed as a building block for future coupling to global coronal models. The model evolves a gravitationally stratified hydrogen plasma with a common velocity and temperature. Equilibrium ionization is obtained from the Saha relation, hydrogen ionization energy is included explicitly in the conserved internal energy, and the density- and temperature-dependent first adiabatic index Γ_1 is supplied by a validated CRASH-derived table. The solver advances the three conserved variables $(\rho, \rho u, \mathcal{E})$ directly, with every carrier quantity derived from the equilibrium closure. Hydrodynamics uses the finite-volume MUSCL–Hancock method with an SWMF-style exact-Riemann Godunov flux, Lie-split from nonlinear backward-Euler physical thermal conduction. The Riemann problem is solved at frozen composition: the equilibrium internal energy splits exactly into a monatomic translational part and an ionization reservoir, and because chromospheric hydrogen ionization relaxes far more slowly than the flow evolves, the reservoir is carried inertly through the face solve while the translational gas responds with $\gamma = 5/3$. The equilibrium adiabatic index Γ_1 remains the acoustic response of the closure and is retained for the reference solver and for diagnostics. The hydrodynamic discretization requires no frozen global hydrostatic reference state: because the Hancock predictor carries the same gravitational source as the corrector and the lower ghost states are a second-order EOS-closed hydrostatic continuation, the initialized stratified column is held at rest to within the float32 round-off scale of the stored state without any residual subtraction. The release mesh has coarse-equivalent $N = 500$, factor-four outer refinement, and 661 actual cells. The lower end of the domain is a mid-chromospheric truncation at 1600 km, not the photosphere. Hydrostatic initialization and boundary states use the same thermodynamic closure, while the upper hydrodynamic ghosts and the fixed 22,000 K conductive datum at the physical outer face are imposed separately. Preliminary release calculations show stable conduction-driven evolution and a coherent upward flow. A 4000 s $N = 500$ calculation has a smooth conserved face mass flux, but no complete matched $N = 1000$ calculation exists at that duration. Long-duration mass-flux convergence below 10% has therefore not been established, so the quantitative evaporation rate remains provisional.

1 Introduction

The chromosphere regulates the exchange of mass and energy between the dense lower solar atmosphere and the corona. Downward electron heat conduction can deposit coronal energy in the transition region and upper chromosphere, increasing pressure and driving an upward enthalpy and mass flux generally identified as chromospheric evaporation (Antiochos and Sturrock, 1978; Fisher et al., 1985). The response depends not only on the deposited energy but also on how that energy is divided among thermal motion, ionization, radiation, and mechanical work.

The transition region is numerically difficult because its temperature gradient occupies a short distance compared with global coronal scales. Under-resolution changes the conductive and enthalpy fluxes and can bias the coronal mass supply (Johnston et al., 2019, 2020). Global wave-driven coronal models consequently use reduced lower-atmosphere treatments rather than resolving the complete dynamic chromosphere along every field line (Sokolov et al., 2021). Such treatments are efficient, but they do not directly evolve the chromospheric mass reservoir under time-dependent conductive forcing.

Thermodynamic closure is especially important in the hydrogen ionization region. A constant ratio of specific heats cannot represent the energy stored in changing ionization state, and a nearly isothermal surrogate does not provide the correct pressure or acoustic response. Moreover, the caloric relation between internal energy and temperature becomes density dependent and nonlinear. Non-equilibrium hydrogen ionization is important in the real chromosphere (Carlsson and Stein, 2002; Leenaarts et al., 2007); the present work adopts equilibrium ionization as a controlled intermediate model that retains the dominant ionization-energy reservoir while avoiding an additional kinetic timescale.

We describe the current `model_column` release: an ionization-aware, field-aligned hydrogen column with finite-volume MUSCL–Hancock hydrodynamics, nonlinear implicit conduction, and local transition-region refinement. The thermodynamic closure is used consistently in initialization, reconstruction, fluxes, conduction, and boundaries. The principal calculation is deliberately limited to a straight prescribed column without magnetic evolution, a self-consistent coronal domain, or an active radiative balance. It is intended as a component for future coupling to AWSoM/SWMF, not as a completed coupled model.

2 Physical Model

2.1 Geometry and assumptions

The independent coordinate s follows a prescribed magnetic field line with strength $B(s)$. Magnetic-flux conservation defines the flux-tube area $A(s)$ through

$$A(s)B(s) = \text{const.} \tag{1}$$

The magnetic field therefore specifies the one-dimensional geometry but is not an evolved variable. In particular, the model has no induction equation or dynamically evolved magnetic-pressure or Lorentz-force term. The plasma contains hydrogen nuclei, electrons, and neutral hydrogen, but all material shares a common field-aligned velocity u and temperature T. Quasi-neutrality gives $n_e = n_p = x n_{\mathrm{H}}$, where $n_{\mathrm{H}} = \rho/m_{\mathrm{H}}$ is the total hydrogen-nucleus density and x is the ionization fraction. The release calculation is therefore a single-fluid equilibrium-mixture model in the strict sense: the solver advances three conserved variables and nothing else.

2.2 Conservation equations

For a time-independent prescribed field and gravitational potential $\Phi(s)$, the active release equations are

$$\frac{\partial \rho}{\partial t} + B\frac{\partial}{\partial s}\left(\frac{\rho u}{B}\right) = 0, \tag{2}$$

$$\frac{\partial(\rho u)}{\partial t} + B\frac{\partial}{\partial s}\left(\frac{\rho u^2 + p}{B}\right) = pB\frac{\partial B^{-1}}{\partial s} - \rho\frac{\partial \Phi}{\partial s}, \tag{3}$$

$$\frac{\partial \mathcal{E}}{\partial t} + B\frac{\partial}{\partial s}\left[\frac{u(\mathcal{E}+p) + q_{\mathrm{c}}}{B}\right] = 0. \tag{4}$$

The conserved state is the three-component vector

$$U = (\rho,\ \rho u,\ \mathcal{E})^{\mathsf{T}}, \tag{5}$$

and the conserved total-energy density and conductive heat flux are

$$\mathcal{E} = e_{\mathrm{int}} + \frac{1}{2}\rho u^2 + \rho\Phi \tag{6}$$

$$q_{\mathrm{c}} = -\kappa_{\mathrm{phys}}(\rho, T)\frac{\partial T}{\partial s}. \tag{7}$$

Because $\rho\Phi$ is included in $\mathcal{E}$, gravitational work is already transported conservatively and no separate gravitational source appears in the energy equation. Equations (2)–(4) are the complete set the solver advances: there is no volumetric heating term, no radiative sink, and no transition-region conductivity modification. Energy enters only through the conductive flux driven by the fixed temperature imposed at the upper physical face.

The reported `model_column` calculation specializes this system to a straight vertical field of constant strength, $B(s) = B_0$ (numerically $B_0 = 1$). Equation (1) then gives constant area, $\partial_s(B^{-1}) = 0$, and Eqs. (2)–(4) reduce to the Cartesian one-dimensional conservation laws. This specialization removes geometric derivatives but does not promote B to a dynamical variable.

2.3 Equilibrium-ionization thermodynamics

The equilibrium ionization fraction follows the pure-hydrogen Saha relation

$$\frac{x^2}{1-x} = \frac{S(T)}{n_\mathrm{H}}, \qquad S(T) = \left(\frac{2\pi m_e k_\mathrm{B} T}{h^2}\right)^{3/2} \exp\left(-\frac{\chi_\mathrm{H}}{k_\mathrm{B}T}\right), \tag{8}$$

with $\chi_\mathrm{H} = 13.6$ eV. The total pressure and internal-energy density are

$$p = (1+x) n_\mathrm{H} k_\mathrm{B} T, \tag{9}$$

$$e_\mathrm{int} = \frac{3}{2} p + x n_\mathrm{H} \chi_\mathrm{H}. \tag{10}$$

The second term in Eq. (10) makes the caloric equation nonlinear. The code recovers temperature from (ρ, e_int) with a safeguarded Newton/bisection inversion and then recomputes x, p, and all derived thermodynamic quantities from the accepted (ρ, T).

Equations (8)–(10) close the system in Eq. (5): the ionization fraction x, the carrier densities $\rho_i = x\rho$ and $\rho_n = (1-x)\rho$, the electron and neutral number densities $n_e = x n_\mathrm{H}$ and $n_\mathrm{HI} = (1-x) n_\mathrm{H}$, and the electron partial pressure $p_e = x n_\mathrm{H} k_\mathrm{B} T$ are single-valued functions of the conserved state. They are diagnostic outputs of the closure, not independently advanced variables, and the solver stores none of them.

At fixed density, the effective volumetric heat capacity is

$$C_V = \left(\frac{\partial e_\mathrm{int}}{\partial T}\right)_\rho = n_\mathrm{H}\left[\frac{3}{2} k_\mathrm{B}(1+x) + \left(\frac{3}{2} k_\mathrm{B} T + \chi_\mathrm{H}\right)\left(\frac{\partial x}{\partial T}\right)_\rho\right]. \tag{11}$$

The ionization derivative can strongly increase C_V through the partial-ionization interval.

Two distinct adiabatic indices are retained. The caloric ratio

$$\gamma_E = 1 + \frac{p}{e_\mathrm{int}} \tag{12}$$

is a diagnostic relation between pressure and total internal energy. It is not the acoustic exponent. The *equilibrium* acoustic response, obtained when the ionization balance re-equilibrates within the disturbance, is

$$\Gamma_1 = \left(\frac{\partial \ln p}{\partial \ln \rho}\right)_s, \qquad c_\mathrm{eq}^2 = \Gamma_1 \frac{p}{\rho}. \tag{13}$$

The release obtains $\Gamma_1(\rho, T)$ from a validated CRASH-derived pure-hydrogen table; in the modelled column its median is 1.090 and it lies below 1.5 across 94.9% of cells, because compression partly ionizes the gas and the ionization reservoir absorbs part of the compressional work. This distinction prevents the ionization energy in e_int from being incorrectly interpreted as an ideal-gas degree of freedom.

2.4 Frozen and equilibrium acoustic limits

Equation (10) splits exactly into a translational term and an ionization reservoir,

$$e_\mathrm{int} = \frac{p}{5/3 - 1} + e_\mathrm{ion}, \qquad e_\mathrm{ion} = x n_\mathrm{H} \chi_\mathrm{H}, \tag{14}$$

so a monatomic gas plus a store of ionization energy is an identity of this closure, not an approximation to it. The two acoustic limits differ only in what happens to e_ion while a compressive disturbance passes. If the composition re-equilibrates within the wave, the response is c_eq above. If it cannot change on that timescale, e_ion is inert, only the translational term responds, and the response is the *frozen* monatomic value

$$c_\mathrm{fr}^2 = \frac{5}{3}\frac{p}{\rho}. \tag{15}$$

Neither exponent is universally correct; the choice is set by the ratio of the ionization/recombination relaxation time to the dynamical time. For chromospheric hydrogen that ratio strongly favours the frozen limit. Carlsson and Stein (2002) find the relaxation time dominated by slow collisional leakage from the ground state, reaching 10^3–10^5 s through the mid-chromosphere and $\sim 10^2$ s at the transition-region base, and remaining 10–10^3 s inside shocks; Leenaarts et al. (2007) confirm this in two-dimensional radiation-MHD and note the energetic consequence, that non-equilibrium hydrogen ionization is a much less effective internal-energy buffer, raising shock temperatures from ≤ 8000 K to 10 000–13 000 K; and Leenaarts (2020) places the chromospheric hydrodynamic timescale at about a minute. The release therefore adopts the frozen limit for the local Riemann interaction (Sect. 3.2), while retaining Γ_1 as the equilibrium derivative of the closure.

2.5 Physical interpretation

Equation (10) makes ionization an internal-energy buffer. Conductive input in a partially ionized layer can increase x substantially while producing a smaller immediate rise in T than a constant-γ gas with the same energy input. The associated changes in pressure, C_V, and Γ_1 modify compressibility and acceleration. Consequently, the mass loaded into an evaporation flow depends on the density-dependent ionization response, not only on the imposed heat flux.

2.6 Thermal conduction

The physical conductivity is the sum of electron and neutral contributions,

$$\kappa_{\rm phys} = \kappa_e + \kappa_n, \tag{16}$$

with the implemented SI closures

$$\kappa_e = \frac{9.2048 \times 10^{-12} n_e T^{5/2}}{n_e + 2.836 \times 10^{-11} n_{\rm HI} T^2}, \tag{17}$$

$$\kappa_n = \frac{0.0342006 n_{\rm HI} T}{1.20613 n_e \sqrt{2T} + 1.70573 n_{\rm HI} \sqrt{T}}. \tag{18}$$

These forms recover electron-dominated conduction as the plasma ionizes and neutral conduction at low ionization (Spitzer, 1962; Vargaftik, 1975; Vranjes and Krstic, 2013). The release uses physical conduction only. At an interior face,

$$q_f = -\kappa_{{\rm phys},f} \frac{T_R - T_L}{\Delta s_f}. \tag{19}$$

The conductivity in Eq. (19) is $\kappa_e + \kappa_n$ and nothing else: the release conduction operator contains no artificial or mesh-scaled diffusivity term and no adaptive transition-region broadening, so physical-conduction-only is a structural property of the discretization rather than a parameter choice.

3 Numerical Method

3.1 Finite-volume hydrodynamics

The hydrodynamic terms in Eqs. (2)–(4) are discretized in conservative finite-volume form on a nonuniform mesh. The authoritative mixture state and hydrodynamic flux are

$$U = \begin{pmatrix} \rho \\ \rho u \\ \mathcal{E} \end{pmatrix}, \qquad F(U) = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ u(\mathcal{E} + p) \end{pmatrix}. \tag{20}$$

The hydrodynamic time discretization is the MUSCL–Hancock method (van Leer, 1979; Toro, 2009; Berthon, 2006). MUSCL supplies the spatial reconstruction: the primitive triple $(\ln \rho, u, \ln p)$ is extrapolated to the two faces of each cell with the asymmetric MC3/Koren limiter (Koren, 1993) at $\beta = 2$, which keeps the reconstructed density and pressure positive; the face temperature then follows by inverting the same equilibrium EOS at the reconstructed density and pressure. The Hancock predictor advances each cell to the half-time level using only its own two extrapolated fluxes together with the same momentum source as the hydrodynamic PDE,

$$U_i^p = U_i^n - \frac{\Delta t}{\Delta s_i}\Big[F\big(W_{i+1/2}^L\big) - F\big(W_{i-1/2}^R\big)\Big] + \Delta t\, S(U_i^n), \qquad S = \big(0,\ -\rho\,\partial_s \Phi + p\,\partial_s \ln A,\ 0\big)^\top, \tag{21}$$

after which the corrector reconstructs from the time average $\frac{1}{2}(W^n + W^p)$ and the Godunov flux below is evaluated on those time-centred face states. Including S in the predictor is essential rather than optional: in a hydrostatic column the pressure-gradient part of the flux difference in Eq. (21) is balanced by gravity, so omitting S leaves the half state with a spurious velocity of order $\Delta t\,|g|$, which the corrector then reconstructs into every face. Because Φ is carried inside $\mathcal{E}$, gravity does no separate work on the energy row and S acts on momentum alone.

3.2 Numerical flux

The release numerical flux is the SWMF-style exact-Riemann Godunov flux (Toro, 2009), evaluated at the frozen composition of Sect. 2.4. At each face the ideal-gas Riemann problem is solved exactly at the fixed monatomic index $\gamma = 5/3$, and the entire non-ideal content of the energy — the Saha ionization energy inside e_{int}, together with the gravitational potential this model carries in $\mathcal{E}$ — is placed in a passive specific offset attached to the material,

$$E_0 = \frac{\mathcal{E} - \frac{1}{2}\rho u^2 - p/(\gamma-1)}{\rho} = \frac{e_{\mathrm{int}} - p/(\gamma-1)}{\rho} + \Phi. \tag{22}$$

Both sides of a face share Φ, so only the equation-of-state content is upwinded. With $(\rho_\star, u_\star, p_\star)$ the exact self-similar solution sampled along $x/t = 0$ and E_0 taken from whichever side the contact declares upwind,

$$\mathcal{E}_\star = \frac{p_\star}{\gamma-1} + \tfrac{1}{2}\rho_\star u_\star^2 + \rho_\star E_0, \qquad F^{\mathrm{God}}_{i+1/2} = \big(\rho_\star u_\star,\ \rho_\star u_\star^2 + p_\star,\ (\mathcal{E}_\star + p_\star)u_\star\big). \tag{23}$$

When the sampled state coincides with a side state this reproduces that side's total energy exactly, and a constant offset is advected without altering (ρ, u, p) at all. Any face whose exact solve fails — inadmissible input, the two-rarefaction vacuum guard, a non-positive star pressure, or an exhausted iteration budget — falls back to the local Lax–Friedrichs (Rusanov) flux on that face alone, and every such event is counted and reported at the end of the run rather than absorbed silently. Both reconstructed states use the same face potential in $\mathcal{E}$, while the explicit gravitational source acts only on momentum, consistently with Eqs. (2)–(4).

Because the release flux propagates the frozen waves of Eq. (15), the acoustic CFL condition is evaluated with the same signal speed, $c^2 = \max(\Gamma_1, 5/3)\,p/\rho$, so that the requested Courant number is the Courant number of the scheme in force. On the release mesh this leaves the timestep unchanged in practice, because the Courant-limiting cell is the fully ionized top cell where $\Gamma_1 = 5/3$ already; the rule matters for other mesh or model shapes, where a partial-ionization cell could become limiting.

A mixture Roe characteristic flux — a local general-EOS linearization of the *equilibrium* system, whose acoustic eigenvalues carry Γ_1 — is retained as a reference solver and is the discrete realization of the competing instantaneous-equilibrium closure. It is not claimed to be an exact nonlinear general-EOS Roe construction satisfying Property U. Selecting it also reverts the Courant condition to c_{eq}, so the comparison remains internally consistent. On the release configuration the two fluxes agree on every conduction-driven observable to a few parts in 10^4, so the choice between them rests on the physical argument of Sect. 2.4 rather than on a measured numerical advantage.

3.3 Hydrostatic balance without a reference state

A conventional discretization leaves a finite mismatch between the pressure-flux gradient and gravity, which can launch flows much larger than the desired evaporation signal. Well-balanced schemes remove this mismatch constructively: either by supplying the intended equilibrium in advance (Berberich et al., 2019), or by reconstructing so that a class of discrete hydrostatic states is preserved to machine precision without prior knowledge of it (Käppeli and Mishra, 2014; Chandrashekar and Klingenberg, 2015). *The present release does neither, and is not a well-balanced scheme in that sense.* It instead removes the two discretization inconsistencies that dominated its own imbalance, and then demonstrates that what remains is negligible at the release resolution.

The first is the predictor source term of Eq. (21). The second is the lower ghost construction: each ghost pressure is now obtained from the trapezoidal, EOS-closed hydrostatic step $p_{k+1} = p_k + \Delta s\,\frac{1}{2}(\rho_k + \rho_{k+1})g$ with $\rho_{k+1} = \rho_{\mathrm{EOS}}(p_{k+1}, T_0)$ iterated to a fixed point, matching the construction already used at the outer face; the first-order form $p_k + \rho_k g \Delta s$ it replaces carried an $\mathcal{O}((\Delta s/H)^2)$ pressure error and hence an $\mathcal{O}(\Delta s/H)$ error in the force balance of the lowest interior cells.

Together these reduce the raw discrete hydrostatic defect of the release configuration, measured as the numerical face mass flux of the initialized $u = 0$ atmosphere, by a factor of about 560, from 7.5×10^{-10} to $1.3 \times 10^{-12}\,\mathrm{kg\,m^{-2}\,s^{-1}}$. That residual is comparable to the estimated float32 round-off scale $\epsilon_{32}\rho\,(|u| + c_s) \approx 1.0 \times 10^{-12}$ of the stored state, and doubling the resolution does not reduce it (2.7×10^{-12} at $N = 1000$), so no clean truncation-error trend survives over the resolutions tested. We do not claim to have identified the residual with any single error source; reconstruction, the EOS inversion, the nonuniform mesh, the boundary closures and float32 storage all contribute at this level. Over 20 s with conduction disabled the initialized column develops $\max|u| = 0.030\,\mathrm{m\,s^{-1}}$ and a relative density drift of 3.5×10^{-5}, against evaporation velocities of 15–40 $\mathrm{m\,s^{-1}}$.

Earlier releases instead froze the initialized equilibrium U_{eq}, evaluated its discrete residual once, and advanced $dU/dt = R(U) - R_{\mathrm{eq}}$, in the manner of schemes that require the equilibrium a priori (Berberich et al., 2019). That construction made one chosen state an exact fixed point, but it bound the solver to a pre-known global equilibrium and continued to subtract a residual evaluated at $t = 0$ long after the transition region had restructured. It is no longer part of the release. Enabling it in the corrected solver changes the 1000 s evaporation solution by 0.4% in the analysis-window velocity and by 0.5% in the peak velocity, which is why it could be removed rather than retained. What has been removed is the *interior* dependence on a global hydrostatic profile; the boundary conditions of Section 4 still capture fixed reservoir data (an outer back pressure and a lower reservoir state) once from the initial condition, as any truncated-domain problem must.

3.4 Implicit nonlinear conduction

Conduction is advanced with backward Euler in temperature. On a general prescribed geometry, the residual discretizes $B\partial_s[(\kappa_{\mathrm{phys}}/B)\partial_s T]$; the reported $B = B_0$ calculation reduces this operator to its constant-area form. The residual contains the nonlinear caloric energy and physical conductivity only. A lagged-conductivity Newton linearization uses the analytic C_V in Eq. (11); each correction requires a tridiagonal Thomas solve. Temperature-domain checks, residual normalization, bounded updates, and stagnation detection guard the solve. The accepted state is written back as $\mathcal{E} = e_{\mathrm{int}}^{n+1} + \frac{1}{2}\rho u^2 + \rho\Phi$ with ρ and ρu untouched, so the stage is conservative in energy and exactly preserves mass and momentum.

3.5 Update sequence

For the release configuration, one step is

$$U^n \xrightarrow{\text{explicit MUSCL–Hancock/Godunov hydrodynamics}} U^* \xrightarrow{\text{implicit physical conduction}} U^{n+1}. \tag{24}$$

Because U is already the physical state of Eq. (5), no intermediate projection or repacking stage is required: Saha equilibrium enters only through the EOS closure that decodes U wherever thermodynamic quantities are needed.

The two stages are composed by first-order Lie splitting: the conduction solve takes its initial temperature from the completed hydrodynamic state U^* and is not symmetrized about it. The MUSCL–Hancock construction supplies second-order space and time centering for the *homogeneous* flux evolution, and the gravitational source enters the predictor so that hydrostatic balance is maintained consistently across both stages; but the corrector evaluates that source at U^n rather than at the half-time state, so the complete sourced hydrodynamic update is not claimed to be formally second-order in time either. With backward-Euler conduction Lie-split onto it, the composite step is *first order in time*, and no second-order temporal claim is made anywhere in this work. This is adequate here because the release timestep is set by the acoustic CFL condition — evaluated, as in Sect. 3.2, with the frozen signal speed of the flux in force — while the physically interesting evolution is far slower, $|u| \ll c_{\mathrm{fr}}$ throughout; raising the temporal order of the hydrodynamics/conduction composition is a separate question that this work does not address.

3.6 Local refinement

The upper transition region is refined statically because it contains the strongest temperature gradient and controls the conductive boundary input. The release mesh is uniform in the lower part of the domain, graded smoothly over approximately 20 km, and four times finer in the upper 53 km. This places fine cells only where the thermal layer requires them.

3.7 Computational implementation

The caloric inversion reuses the previous cell temperature when valid and falls back to a safeguarded bracketed solve. Cell-local EOS, reconstruction, and conduction assembly are parallelized with deterministic OpenMP ordering; the tridiagonal conduction solve remains serial. The production launcher uses 12 OpenMP threads and CFL = 0.50 for the validated $N = 500$/R4, 661-cell release configuration. The conservative hard-coded solver default remains 0.25.

4 Model-Column Configuration

4.1 Domain, initial atmosphere, and grid

The principal domain covers heights 1600–2153 km, a length of 553 km. Gravity is vertical and the prescribed field line is straight. The lower end of this interval is a deliberate truncation of the modeled domain rather than a physical surface: it is not the photosphere, but the height above which the Saha/LTE closure of Section 2.3 is intended to be used. The chromosphere below 1600 km is real but outside the model, and no claim is made that this closure describes it. The initial temperature is a shape-preserving PCHIP interpolation of Model C7 (Avrett and Loeser, 2008). Ionization is recomputed from Eq. (8), after which pressure and density are reintegrated hydrostatically with the same EOS used during evolution. The initial velocity is zero.

The release mesh has coarse-equivalent $N = 500$. Fourfold refinement begins 500 km above the lower boundary; a smooth grade occupies approximately 2080–2100 km and the fine plateau covers approximately 2100–2153 km. Refinement increases the actual count to 661 cells, with coarse and fine spacings of approximately 1.105 and 0.276 km, respectively. Figure 1 shows the exact mesh and initial temperature used by this configuration.

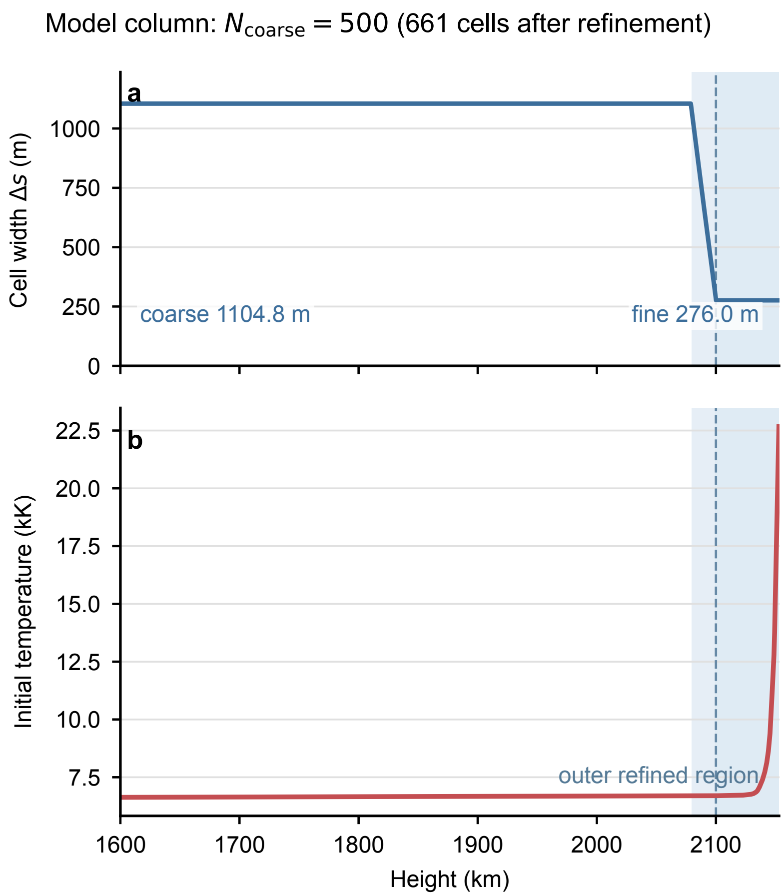

Figure 1: Release mesh and EOS-aware initial temperature for the coarse-equivalent $N = 500$ column. The mesh is reconstructed with the same static outer-refinement algorithm used by `model_column`.

4.2 Boundary conditions

Table 1 summarizes the independent physical data and the quantities derived from them. The two hydrodynamic ghosts at each end are a numerical closure for the MUSCL reconstruction stencil, not an extension of the physical model beyond the domain; at the lower end in particular they do not assert that the Saha/LTE description continues below 1600 km. The conductive datum is separate and exists at the physical outer face.

The physical content of the lower boundary is a single statement: on the timescales resolved here, the chromosphere below the truncation behaves as a quasi-static stratified reservoir at the truncation temperature, so the column neither drains into it nor is driven by it. Whether this is too rigid a statement was tested directly by replacing it with a gravity-aware characteristic condition that imposes only the incoming upward acoustic invariant and lets outgoing disturbances leave. Over 20 s hydrostatic, 100 s and 1000 s conduction-driven runs that change left the hydrostatic residual unchanged, altered the evaporation-window velocity by at most 0.4%, and left the detrended oscillation content of the base region unchanged. The disturbance that reaches 1600 km in this model is a slow quasi-static pressure adjustment rather than acoustic ringing, so the fixed reservoir is not an important reflector here and the simpler condition was kept. One difference is not negligible and should not be read out of that null result: over 1000 s the characteristic condition admitted mass from below and recovered about 17% of the column mass loss. The fixed reservoir is a statement that no mass crosses 1600 km, which constrains the mass budget even where it leaves the velocity field unchanged.

Table 1: Release boundary conditions. Derived quantities are recomputed through the equilibrium EOS rather than imposed independently.

Quantity	Lower boundary (1600 km truncation)	Upper boundary
Velocity u	Quasi-static reservoir, $u = 0$	Live top-cell velocity, capped by $\lvert u\rvert \leq 0.1c_s$
Pressure p	Trapezoidal EOS-closed discrete-HSE continuation through both ghosts	First ghost fixed to the external back pressure; second ghost from one-sided HSE
Hydrodynamic temperature T_{hydro}	Fixed reservoir temperature	Zero-gradient from the live top cell in both hydro ghosts
Density ρ	Derived from ghost pressure and fixed T through the equilibrium EOS	Derived from ghost pressure and T_{hydro} through the equilibrium EOS
Saha fraction x	Derived from the resulting (ρ, T) state	Derived from the resulting $(\rho, T_{\mathrm{hydro}})$ state
Conductive temperature	Fixed lower reservoir temperature	$T_{\mathrm{face}} = 22{,}000$ K at the physical outer face
Thermodynamic/total energy	Packed consistently from the reservoir state	Packed from the EOS-derived hydro ghost state; not independently prescribed

For the upper conductive flux, the distance from the top-cell center to the Dirichlet datum is

$$d_{\mathrm{boundary}} = \frac{1}{2}\Delta s_{\mathrm{top}}, \qquad q_{\mathrm{top}} = \kappa_{\mathrm{face}} \frac{T_{\mathrm{face}} - T_{\mathrm{top}}}{d_{\mathrm{boundary}}}, \tag{25}$$

where κ_{face} is evaluated from the live top-cell state and a Saha-equilibrium physical-face state constructed from the external pressure and T_{face}. This temperature is not the hydro ghost temperature, and the second hydro ghost has no conductive role.

Pressure, temperature, density, and energy are therefore not all imposed independently at the upper hydrodynamic boundary. The single external hyperbolic datum is the reservoir back pressure; velocity and hydrodynamic temperature are extrapolated from the live solution, and density, composition, and energy follow from the EOS.

4.3 Reproducibility settings

Table 2: Principal release configuration.

Quantity	Setting
Height interval	1600–2153 km
Geometry	straight, constant cross section, vertical gravity
Thermodynamics	Saha equilibrium H; ionization energy; tabulated Γ_1
Initial atmosphere	Model C7 PCHIP temperature; Saha-consistent HSE
Hydrodynamics	MUSCL–Hancock on $(\ln\rho, u, \ln p)$; MC3/Koren limiter, $\beta = 2$; SWMF exact-Riemann Godunov flux at frozen $\gamma = 5/3$ on the time-centred face states
Time integration	Lie split; MUSCL–Hancock hydrodynamics (source evaluated at U^n) + backward-Euler conduction; first order overall
Well balancing	none: reference-free, no frozen equilibrium state
Conduction	nonlinear backward Euler; electron + neutral physical conductivity
Numerical conduction	absent from the operator; physical conduction only
Lower boundary	1600 km truncation; quasi-static reservoir, $u = 0$, trapezoidal EOS-closed HSE ghosts, conductive Dirichlet
Upper hydro boundary	fixed back pressure, HSE ghosts, zero-gradient T, Mach cap 0.1
Upper conduction boundary	fixed $T_{\mathrm{face}} = 22{,}000$ K at the physical face; $d = \Delta s_{\mathrm{top}}/2$
Grid	coarse-equivalent $N = 500$, factor-four upper refinement, 661 cells
Conserved state	$(\rho, \rho u, \mathcal{E})$; carrier quantities derived from the EOS
Active sources	physical conduction only; the update has no other stage
Production launcher CFL	0.50, on the frozen signal speed (hard-coded fallback 0.25)
Production OpenMP configuration	12 threads; deterministic execution

5 Verification

5.1 Thermodynamic verification

The Saha implementation is tested from nearly neutral to nearly fully ionized limits and reproduces Eq. (8) to a relative tolerance of 2×10^{-13} in double precision. Caloric round trips recover temperature after evaluating $e_{\text{int}}(\rho, T)$ and inverting it, including states in the partial-ionization interval. The production Γ_1 table is validated against independent CRASH-derived values; the worst relative difference in the audited comparison is 3.1×10^{-4}. Independent isentropic density perturbations at ionization fractions near 0.1, 0.5, and 0.9 recover the tabulated acoustic speed to within 5×10^{-4} relative.

5.2 Hydrostatic and boundary verification

The C7/PCHIP state satisfies the discrete Saha hydrostatic relation cell by cell, and all four ghost states decode to the same equilibrium closure. A 50 s undriven test of the decoupled upper boundary retained a maximum speed of 0.0306 m s^{-1} and a relative mass change of 3.08×10^{-8}. Focused tests also verify that changing the conductive wall temperature alters the conduction update without changing the hydrodynamic ghost state, while changing the live top temperature updates the hydro ghost without changing the wall.

5.3 Conduction verification

Closed-boundary conduction tests on nonuniform grids conserve the grid-weighted total energy to within 3×10^{-6} relative while smoothing strong temperature gradients. The accepted nonlinear state satisfies independent backward-Euler residual checks below 5×10^{-5} in the strongest test. The production diagnostics reconstruct the outer physical and numerical face conductivities and recover the solver heat flux to floating-point accuracy.

5.4 Release-numerics verification

The static mesh generator is tested for exact domain closure, monotone cell-center locations, adjacent-width control, and consistency of the nonuniform finite-volume metrics. A defaults-only $N = 500$ release smoke test reproduces an explicitly selected Godunov/$\ln p$ calculation bitwise, and requires that the exact Riemann solve supplied every one of the 2.4×10^7 faces with no fallback. At 100 s it has a maximum reconstructed face-pressure mismatch of approximately 2×10^{-4}%, face-flux roughness $Q_{\text{eff}} = 6.9 \times 10^{-4}$, and no negative velocity cells. Controlled comparisons show that pressure reconstruction reduces the face-pressure mismatch by two to three orders of magnitude under both Roe and Rusanov. The exact Riemann solver itself is verified against the five standard reference problems of Toro (2009) to the precision their star states are tabulated to, and a shock-tube test confirms that the energy offset of Eq. (22) is inert to the hydrodynamics. The release-validation script also exercises the face-local Rusanov fallback and the physical-face conduction boundary.

6 Results

6.1 Thermodynamic evolution and ionization-energy buffering

The fixed physical-face temperature conducts energy into the initially cooler upper chromosphere. The nonlinear EOS partitions the resulting internal-energy increase between translational energy and hydrogen ionization energy. As the ionization fraction rises, the enhanced effective heat capacity moderates the temperature increase and changes the pressure response. No quantitative energy partition is claimed because the current saved diagnostics do not separately integrate all caloric, ionization, and mechanical contributions over the domain.

6.2 Release evaporation response

The current $N = 500$/R4 calculation carried to 4000 s (measured with the Roe reference flux) uses the release reconstruction, flux, physical-face boundary, and physical conduction only. Over the upper-chromosphere diagnostic window it gives a mean effective mass flux of approximately 3.57×10^{-10} kg m^{-2} s^{-1}, with velocity between about 3 and 11.6 m s^{-1}. The normalized roughness measures are $Q_V \approx 0.0031$ and $Q_M \approx 0.0019$, and the maximum reconstructed face-pressure mismatch is approximately 0.0023%. These diagnostics support a coherent upward transport signal rather than a boundary-cell artifact.

6.3 Resolution limitation

The numerical-method decision is stronger than the current resolution claim. Earlier controlled comparisons established that the $N = 500$ release-shaped mesh resolves the physical conductive layer and that its short-to-intermediate-time conductive input is close to $N = 1000$. For the $\ln p$ method, however, the matched $N = 1000$ run at 4000 s was stopped at 2080 s. The $N = 500$ long-duration evaporation mass flux is therefore not established as grid-converged to better than 10%. No formal convergence order or limiting evaporation rate is reported.

7 Discussion

The principal physical result is that equilibrium ionization changes the relation between deposited conductive energy and hydrodynamic response. A constant-γ gas would map internal energy to pressure with a density-independent ratio, whereas the present closure can store energy in ionization and lower both the immediate temperature rise and the equilibrium acoustic exponent — an effect that is itself suppressed when the ionization state cannot follow the disturbance (Sect. 2.4). A nearly isothermal surrogate may suppress temperature variation but does not reproduce this density-dependent caloric reservoir or the associated compressibility. Applying one closure consistently across hydrodynamics, conduction, initialization, reconstruction, and boundaries is therefore essential: using ionization energy only in one stage would create incompatible temperatures and pressure forces.

The coherent upward mass flux demonstrates the intended mechanism for a dynamic lower boundary. In a future coupled calculation, a coronal model could supply downward heat flux and back pressure while the column returns time-dependent mass, enthalpy, and thermodynamic states along field lines. The present result establishes the numerical and thermodynamic component needed for that exchange, but no AWSoM/SWMF coupling has yet been implemented.

Several limitations define the current scope. The most consequential is the treatment of ionization. The release freezes the composition inside the local Riemann problem, for the relaxation-time reasons of Sect. 2.4, but the equation of state still decodes every state through instantaneous Saha equilibrium, so the composition is reset to equilibrium at the end of every timestep: the algorithm is effectively an equilibrated state, a frozen-composition hydrodynamic step, and an equilibrated state again. These are separate modelling choices, and the first does not imply the second. Read as an operator-split instantaneous-relaxation approximation, the second is justified only when the relaxation time is short — the opposite of the regime that justifies the first. The relevant comparison is the physical relaxation time against the physical dynamical time, not against the numerical timestep, and on that comparison chromospheric hydrogen fails the instantaneous-equilibrium assumption by one to three orders of magnitude (Carlsson and Stein, 2002; Leenaarts et al., 2007); Leenaarts and Wedemeyer-Böhm (2006) find the equilibrium ionization degree varying by more than twenty orders of magnitude between shocked and unshocked gas where the actual non-equilibrium degree is nearly constant. The equilibrium decode is therefore a known approximation of unquantified error, most favourable in exactly the smooth, subsonic, slowly varying regime the release reports and least favourable in the strongly compressive regime it does not claim. A physically more complete model would evolve the ionization fraction with finite-rate ionization and recombination rather than enforcing Saha every step; that is a modelling project, not a correction, and it is not attempted here. The model is one-dimensional with prescribed geometry and no magnetic evolution. The principal calculation does not include a self-consistent coronal domain, and radiative balance is incomplete because the release model carries no radiative sink; adding one, together with an adaptive transition-region conduction treatment, is the natural next step and would require its own convergence campaign. The upper atmosphere is represented by fixed back pressure and a fixed physical-face conductive temperature. The lower truncation is closed with a $u = 0$ reservoir, which suppresses replenishment from the unresolved chromosphere below 1600 km; this matters more for the mass budget than for the flow. The characteristic-boundary experiment of Section 4 changed the evaporation velocity by less than 0.4%, but recovered approximately 17% of the 1000 s column mass loss, so the choice of lower closure is not negligible for a mass-loading interpretation even though it is nearly irrelevant to the velocity field and to the oscillation content. A complete long-duration resolution comparison is not yet available. These restrictions make the present calculation a controlled release demonstration, not a quantitative prediction of a solar evaporation rate.

8 Conclusions

We have implemented a thermodynamically consistent equilibrium-ionization closure for a field-aligned chromospheric column. The solver advances the three conserved variables $(\rho, \rho u, \mathcal{E})$ directly, and Saha ionization, pressure, hydrogen ionization energy, effective heat capacity, and the CRASH-derived Γ_1 are used consistently in hydrodynamics, implicit physical conduction, hydrostatic initialization, reconstruction, and boundaries. The release uses $(\ln \rho, u, \ln p)$ MUSCL reconstruction inside a MUSCL–Hancock step whose predictor carries the same gravitational source as the corrector, an SWMF-style exact-Riemann Godunov flux at frozen composition on the time-centred face states, a 661-cell $N = 500/\mathrm{R4}$ mesh, and a 22,000 K conductive datum at the physical outer face. The hydrodynamic operator requires no frozen global hydrostatic reference state; the initialized column is held at rest with a residual face mass flux comparable to the float32 round-off scale of the stored state. The boundary conditions still hold fixed reservoir data captured once from the initial condition. Conductive forcing produces a coherent upward flow, but a complete matched long-duration resolution comparison for the release method remains outstanding. The limiting quantitative evaporation rate is therefore provisional.

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